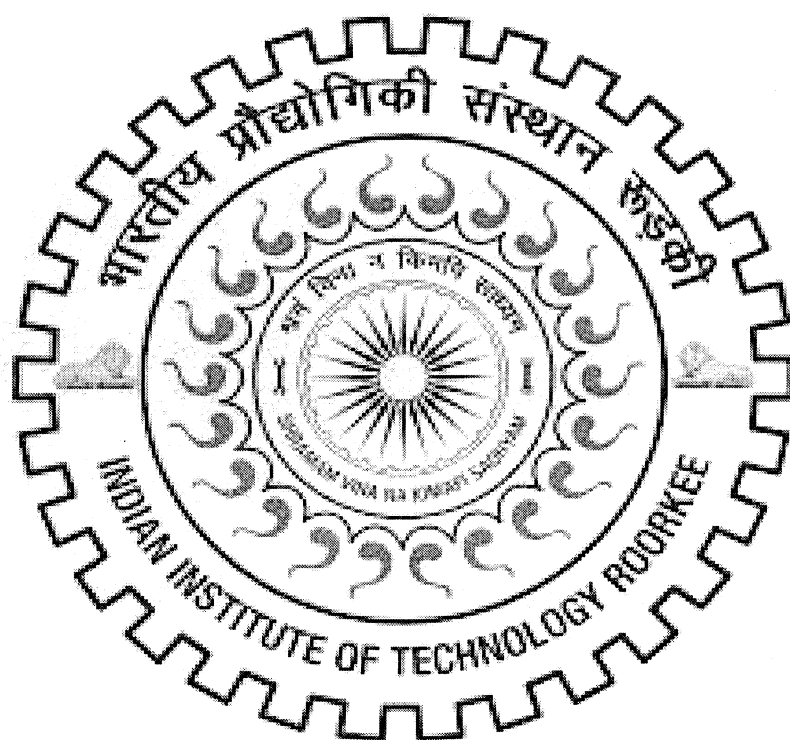


INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF PHYSICS

(ANALOG ELECTRONICS LAB)



LAB MANUAL

CHARACTERISTICS OF MOSFET

N-CHANNEL MOSFET DRAIN AND TRANSFER CHARACTERISTICS

AIM: - To study Transfer and Drain Characteristics of an N-Channel MOSFET in Common N Source Configuration.

APPARATUS: - MOSFET (IRF 540), Bread Board, Resistor, Connecting Wires, Ammeter (mA), DC Power Supply (0-20V), Voltmeter (V).

PROCEDURE:

DRAIN CHARACTERISTICS:

1. Connect the circuit as per given diagram properly.
2. Keep V_{GS} constant at some value say 4V by varying V_{GG} .
3. Initially Vary V_{DS} in steps of 0.2V, once current gets constant then vary V_{DS} in step of 1V up to 15 volts and measure the drain current I_D . Tabulate all the readings.
4. Repeat the above procedure for V_{GS} as 5V, 6V and 7V.
5. Plot the output characteristics V_{DS} vs I_D .

TRANSFER CHARACTERISTICS:

1. Connect the circuit as per given diagram properly.
2. Set the voltage V_{DS} constant at 8V by varying V_{DD} .
3. Vary V_{GS} by varying V_{GG} in the steps of 0.5V from 0V to 10V and note down value of drain current I_D . Tabulate all the readings.
4. Plot the transfer characteristics V_{GS} vs I_D .

OBSERVATION TABLE:

DRAIN CHARACTERISTICS

$V_{GS} = 4\text{ V}$		$V_{GS} = 5\text{ V}$		$V_{GS} = 6\text{ V}$		$V_{GS} = 7\text{ V}$	
$V_{DS}\text{ (V)}$	$I_D\text{ (mA)}$	$V_{DS}\text{ (V)}$	$I_D\text{ (mA)}$	$V_{DS}\text{ (V)}$	$I_D\text{ (mA)}$	$V_{DS}\text{ (V)}$	$I_D\text{ (mA)}$

TRANSFER CHARACTERISTICS

VDS = 8 V	
VGS (V)	ID (mA)
0	
0.5	
1.0	
1.5	
.	
.	
.	
.	
.	
.	
10	

CALCULATION:

1. **Threshold voltage V_T** : Gate to source voltage at which, drain current starts flowing.
2. **Transconductance g_m** : Ratio of small change in drain current (ΔI_D) to the corresponding change in gate to source voltage (ΔV_{GS}) for a constant V_{DS} .

$$g_m = \Delta I_D / \Delta V_{GS} \text{ at constant } V_{DS}$$

3. **Output drain resistance** : It is given by the relation of small change in drain to source voltage (ΔV_{DS}) to the corresponding change in Drain Current (ΔI_D) for a constant V_{GS} .

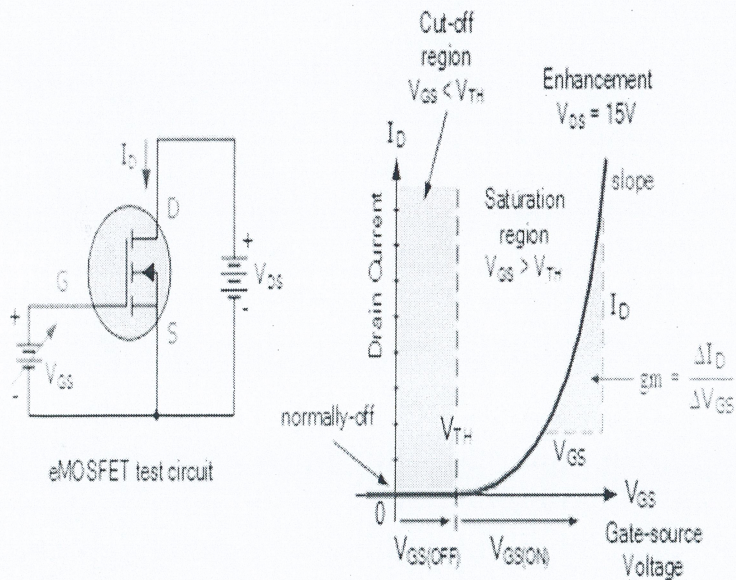
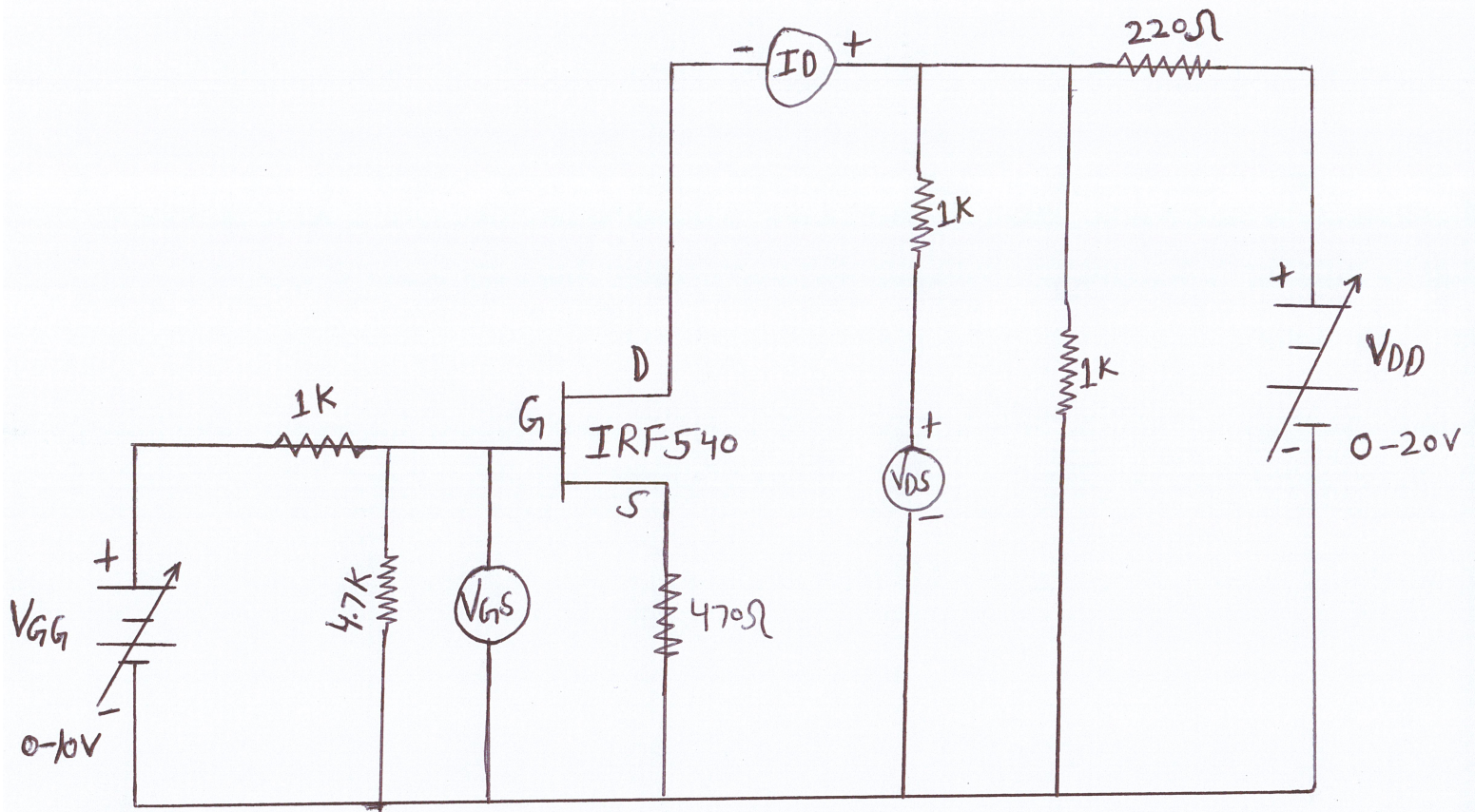
$$r_d \text{ or } r_o = \Delta V_{DS} / \Delta I_D \text{ at a constant } V_{GS}$$

RESULT:

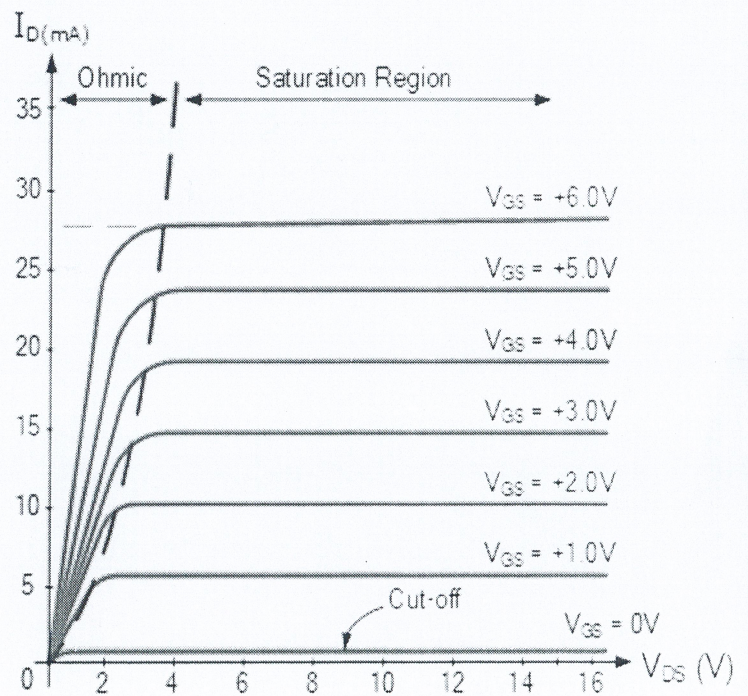
1. V_T : _____
2. g_m : _____
3. r_o : _____

CONCLUSION:

CIRCUIT DIAGRAM -



Transfer characteristics



Drain Characteristics